

HFD 105:

Image File Formats - JPG, TIF, PNG, GIF, PDF

The most common image file formats, the most important for cameras, printing, scanning, and internet use, are JPG, TIF, PNG, and GIF.

Digital cameras and web pages normally use **JPG/ JPEG** files stands for **Joint Photographic Experts Group** - because JPG heroically compresses the data to be very **much smaller in the file**.

- However JPG uses lossy compression to accomplish this feat, which is a strong downside.
- In general today, JPG is rather unique in this regard, using lossy compression allowing very small files of lower quality, whereas almost any other file type is lossless (and larger).
- JPG has small file sizes, making it ideal for web use, photography, and storage.

TIF (Tag Image File) is lossless (including LZW compression option), which is considered the highest quality format for commercial work. (*Lossless refers to data compression allowing the original data to be perfectly reconstructed, with zero loss of information or quality.*)

The TIF format is not necessarily any "higher quality" - (the image pixels are what they are), and most formats other than JPG are lossless too.

- And TIF is the most versatile, except that web pages don't show TIF files. For other purposes however, TIF does most of anything you might want in RGB, CMYK, LAB, or Indexed color Colour profiles.

2. **GIF (Graphics Interchange Format.)**- basic images that appear mainly on the internet. Was designed by CompuServe in the early days of computer 8-bit video, before JPG, for video display at dial up modem speeds. GIF always uses lossless LZW compression,

But GIF is still very good for web graphics (i.e., with a limited number of colors). For graphics of only a few colors, GIF can be much smaller than JPG, with more clear pure colors than JPG).

3. **PNG (Portable Network Graphics)** is a file format widely used for web graphics, logos, and icons . Can replace GIF today (web browsers show both), and PNG also offers many options of TIF.

- PNG was invented more recently than the others, designed to bypass possible LZW compression patent issues with GIF, and since it was more modern, it offers other options
 - One additional feature of PNG is transparency for 24 bit RGB images.
 - Normally PNG files are a little smaller than LZW compression in TIF or GIF.
4. Camera **RAW** files are very important of course, but RAW files must be processed to regular formats (JPG, TIF, etc) to be viewable and usable in any way.

Best file types for these general purposes:

	Photographic Images	Graphics, including Logos or Line art
Properties	Photos are continuous tones, 24-bit color or 8-bit Gray, no text, few lines and edges	Graphics are often solid colors, with few colors, up to 256 colors, with text or lines and sharp edges
For Unquestionable Best Quality	TIF or PNG (lossless compression)	PNG or TIF (lossless compression,)
Smallest File Size	JPG with a higher Quality factor can be decent.	TIF or GIF or PNG (graphics/logos without gradients normally permit indexed color of 2 to 16 colors for smallest file size)
Maximum Compatibility (PC, Mac)	TIF or JPG	TIF or GIF
Worst Choice	256 color GIF is very limited color, and is a larger file than 24 -bit JPG	JPG compression adds artifacts, smears text and lines and edges

Major **considerations** to choose the necessary file type include:

- **Compression quality** - Lossy for smallest files (JPG), or Lossless for best quality images (TIF, PNG).
- **Full RGB color for photos** (TIF, PNG, JPG), or Indexed Color for graphics (PNG, GIF, TIF).

- **16-bit color (48-bit RGB data)** is sometimes desired (TIF and PNG).
- **Transparency or Animation** is used in graphics (GIF and PNG).
- **Documents** - line art, multi-page, text, fax, etc - this will be TIF.
- **CMYK color** is certainly important for commercial prepress (TIF).

Difference in photo and graphics images

Photo images have **continuous tones**, meaning that adjacent pixels often have very similar colors, for example, a blue sky might have many shades of blue in it.

Graphic images are normally not continuous tone (gradients are possible in graphics, but are seen less often). Graphics are drawings, not photos, and they use relatively few colors, maybe only two or three, often less than 16 colors in the entire image.

In a color graphic cartoon, the entire sky will be only one shade of blue where a photo might have dozens of shades

JPG files are very small files for continuous tone photo images, but JPG is poor for graphics, without a high Quality setting.

5. PDF (Portable Document Format)

PDF (**Portable Document Format**) is a file format that has captured all the elements of a printed document as an electronic image that you can view, navigate, print, or forward to someone else.

PDF files are created using Adobe [Acrobat](#) , Acrobat Capture, or similar products.

To view and use the files, you need the free Acrobat Reader, which you can easily download. Once you've downloaded the Reader, it will start automatically whenever you want to look at a PDF file.

PDF files are especially useful for documents such as magazine articles, product designs, or designwork generated from CAD application softwares in which you want to preserve the original graphic appearance online.

A PDF file contains one or more page images, each of which you can zoom in on or out from. You can page forward and backward.

PDF makes it easy to share ideas and information

Reliable — When you use PDF files to communicate, you don't have to worry about whether people have the same software you used to create your originals.

Open standard — The integrity and longevity of the more than one billion PDF files in existence today

Trusted — Businesses and government agencies around the world use PDF as a standard format for exchanging documents more securely. You can, use security features, including password protection, you can help prevent people from accessing, copying, editing, or printing PDF documents.

You can also permanently delete sensitive information from PDFs by redacting visible text and illustrations or removing hidden information.

Multiplatform — You can view and interact with PDFs on virtually any platform, including Windows®, Mac OS, and mobile platforms including Android™ and iOS for iPhone and iPad.

Rich in file integrity — PDFs look just like the originals and preserve all the source file information, even when text, drawings, videos, audio, 3D maps, full-color graphics, photos, and business logic are combined in a single file or PDF Portfolio.

Simple to sign — Virtually anyone can electronically sign PDFs using free Adobe Reader XI software or Adobe Reader for mobile. PDF files can support basic signature images as well as certificate-based signatures that can be verified by independent third-party services.

Searchable — When you need to find a particular word or phrase in a PDF, it's easy to search text and metadata, including scanned text that was converted using optical character recognition (OCR) technology.

Accessible — Over time, the PDF specification has been expanded to work with assistive technologies, making PDF files more accessible to people with disabilities such as low vision or blindness.

Broadly adopted — Whether your business needs a specialized solution for integrating PDF files into daily workflows — or you'd like to receive training or consultation on the most efficient use of PDFs — support is available. Thousands of vendors worldwide offer PDF-based solutions, plug-ins, and tools, as well as PDF-focused training and consulting.

How to save in image file formats:

1. CorelDraw.
 - A) Use the following steps: FILE – SAVE AS – SAVE AS TYPE (gives options AI , CDR etc.) this options enables you to save image as CDR-Coreldraw (image that can be opened via coreldraw software only) & AI – (Adobe Illustrator that can be opened through Adobe Illustrator and CorelDraw softwares)
 - B) Use the following steps: FILE – EXPORT – SAVE AS TYPE (gives options AI ,PNG, TIF, JPEG)
 - C) Use the following steps: FILE – PUBLISH TO PDF – SAVE AS TYPE (pdf)
2. Adobe Illustrator
 - A) Use the following steps: FILE – SAVE AS – SAVE AS TYPE (gives options AI and PDF)
 - B) Use the following steps: FILE – EXPORT – SAVE AS TYPE (gives options ,PNG, TIF, JPEG)

How to get/import image file formats to a design page.

1. CorelDraw.

Use the following steps: FILE – IMPORT – FILE NAME (select image that is already saved in the computer –eg in a folder or on the Desktop)- IMPORT

2. Adobe Illustrator

Use the following steps: FILE – PLACE – FILE NAME (select image that is already saved in the computer –eg in a folder or on the Desktop)- PLACE

COMPUTER-AIDED DESIGN (CAD)

Computer-Aided Design

The term computer-aided (CAD) design includes all the computer applications and hardware devices that can be used to aid digital design.

CAD speeds up the design process by making it quick and easy to test and modify ideas before production starts. This reduces mistakes and cuts costs.

Uses of CAD in textiles design include:

1. Graphics applications enable ease the production and storage of accurate working drawings and lay plans.
2. Colourways can be accurately modelled at the design stage.
3. Material quantities and costs can be easily calculated.
4. Computer networks improve communication between designers, clients and manufacturers thus speeding up the design-feedback loop
5. Compatibility/used with many other applications, including programs for painting, scanning, processing, editing, publishing, and analyzing images.
6. Simulate virtual products in 3D, (3 DIMENSIONAL)from which clients can choose one to be sampled in fabric. This saves the time and cost of sampling a large selection of real products.

General CAD

Main CAD softwares : CorelDraw, CorelPhotopaint, CorelTrace & Adobe Creative suite (which includes Photoshop, Illustrator, InDesign, Dream Weaver).

Colour scanners and printers and A4 graphic tablets enabling users to upload or draw their own images, manipulate them and print them out.

Textile Computer-aided design (CAD) facilities use specific software relevant to their discipline choice in addition to general CAD software. Examples:

- Knit CAD
- Weave CAD
- Fashion CAD
- Print CAD

Knit CAD

In addition to the generic CAD software applicable to all designers are specialist CAD packages for knitwear designers, including:

Weave CAD

Specialist ‘Scotweave’ software that enables the development of complex weave structures. The software simulates the structure and also produces peg plans and lifting plans for the designer simplifying the weave process and helping to guarantee the intended result.

Fashion CAD

The fashion CAD suite is specifically dedicated to fashion design, pattern making, lay planning and fashion production. Examples

- Lectra, Kaledo Style illustration software
- Modaris pattern drafting, modification and grading software
- Diamino marker planning and fabric utilisation software

Print CAD

Most designers routinely use Adobe Photoshop to develop their images. These images may then be fed directly to the digital printer if appropriate. When the designer wishes to screen print their images (or part of their image) they will often use the 'studio master' software for separating and preparing their screens.

What is CAD/CAM?

CAD/CAM (Computer-Aided Design and Computer-Aided Manufacturing) refers to computer software that is used to both design and manufacture products.

Computer-aided manufacturing (CAM) is the use of a wide range of Product Lifecycle Management computer-based software tools that assist engineers, in the manufacture of product components.

3D models of components generated in CAD software are used to generate CNC code to drive numerical controlled machine tools.

CAD is the use of computer technology for design and design documentation.

CAM software uses the models and assemblies created in CAD software to generate tool paths that drive the machines that turn the designs into physical parts.

CAD/CAM software is most often used for machining of prototypes and finished parts.

Computer-aided manufacturing (CAM) machines:

In the textile industry, machines have increased efficiency and production rates dramatically.

In the past, workers had to sit at tables making everything by hand, stitch by stitch, but now machines are now doing the same work much faster.

Whether it's knitting, zipper-making, sewing, or fabric making, there is a device for virtually every aspect of textile creation.

Knitting Machines

The act of knitting involves creating a series of loops out of yarn.

When done by hand, this can take hours for one piece. Changing the yarn colors, adding new yarn when one ball runs out, and other necessary actions make hand knitting clothing largely inefficient for mass production.

Knitting machines mimic the same motions made by hand knitting, but on a large scale, reducing the amount of time it takes to make each piece of clothing and improving productivity.

Lace Making Machines

True lace is woven, braided, or looped thread created into beautiful designs independent of a backing fabric.

Originally, lace making was a needlecraft and was handmade by seamstresses.

Since this art is painstaking in its handmade form, lace garments can oftentimes take days to make.

Lace making machines utilize sewing and knitting motions, as well as crochet, to create pieces in a fraction of the time it would take one person to make by hand.

Quilt Making Machines

While many quilters still exist today and continue to make handmade pieces, this craft has largely been turned over to technology.

Quilts consist of three layers of fabric, generally: the top fabric; batting, or insulation; and backing.

While this can make for a very slow project when done by hand, machines have combined sewing and other industrial equipment to create quilts in almost no time at all.

Mass-produced bedspreads, duvet covers, and similar items are made by quilt making devices.

Label Making Machines

Textile products, like clothing, almost always include a label to express to the customer who made the item, meaning the brand name, and sometimes the material of the product, as well.

Before label-making devices came to be, each label would have to be hand stitched onto the fabric, which could take several minutes to an hour, depending on the size of the label.

The label itself, which is usually made from a ribbon-like material or other fabric, would also have to be made.

Now, the process of making the label, embroidering it with branding and material information, and affixing it to the product is all done with one machine capable of labeling hundreds of products an hour.

Textile Bleaching Machines

Oftentimes, fabric in its raw form, such as just-woven yarn, is brown or inconsistent in color. Whether or not the fabric is supposed to be white, it may still need to be bleached in order for the fabric dye to take in a consistent manner.

Dealing with large amounts of bleach by hand can be hazardous to a worker's health, and can also lead to bleaching that is not uniform.

Bleaching machines treat fabric in a uniform and consistent way, and then dry and clean the fabric properly before it goes on to become the product it is intended to be.

Textile machines have helped in the mass production of a variety of products. However, some would argue that they have done so at the expense of compromising the true art form they mimic.

For instance, hand knitted garments are likely to be of a much higher quality than those made by a machine. They do take longer to make and are more expensive to buy, but the quality of the garment and the integrity of the craft is not sacrificed.

However, the mass market will continue to benefit from technological efficiency.

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